7. Run the modified version of PoI where the minimum improvement threshold is set at 0.1. Replace the line that initializes the BayesOpt policy with:

```
policy = botorch.acquisition.analytic.ProbabilityOfImprovement(
    model, best_f=train_y.max() + 0.1
)
```

This policy is more explorative and outperforms regular PoI. Figure A.7 shows the progress of this policy at iteration 17, where it first achieves an accuracy of at least 90 percent.

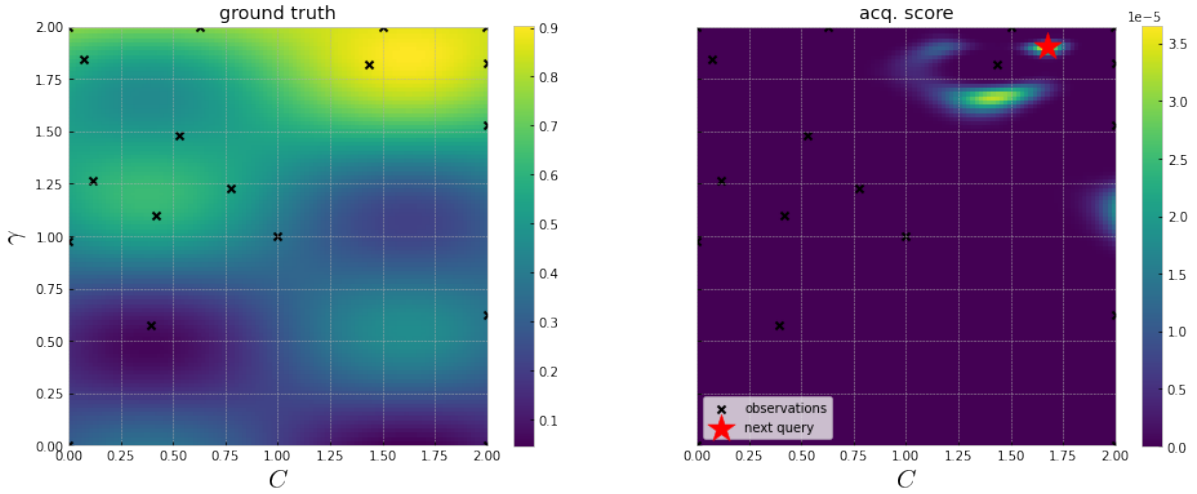


Figure A.7 Optimization progress of the modified Probability of Improvement at iteration 17, where the policy first achieves an accuracy of at least 90 percent.

Here, $C = 1.6770$ and $\gamma = 1.9039$ are the parameters giving this accuracy.

8. Run the Expected Improvement policy on this objective function. Replace the line that initializes the BayesOpt policy with:

```
policy = botorch.acquisition.analytic.ExpectedImprovement(
    model, best_f=train_y.max()
)
```

This policy performs well on our objective function, finding an accuracy of at least 90 percent at iteration 4, as shown in

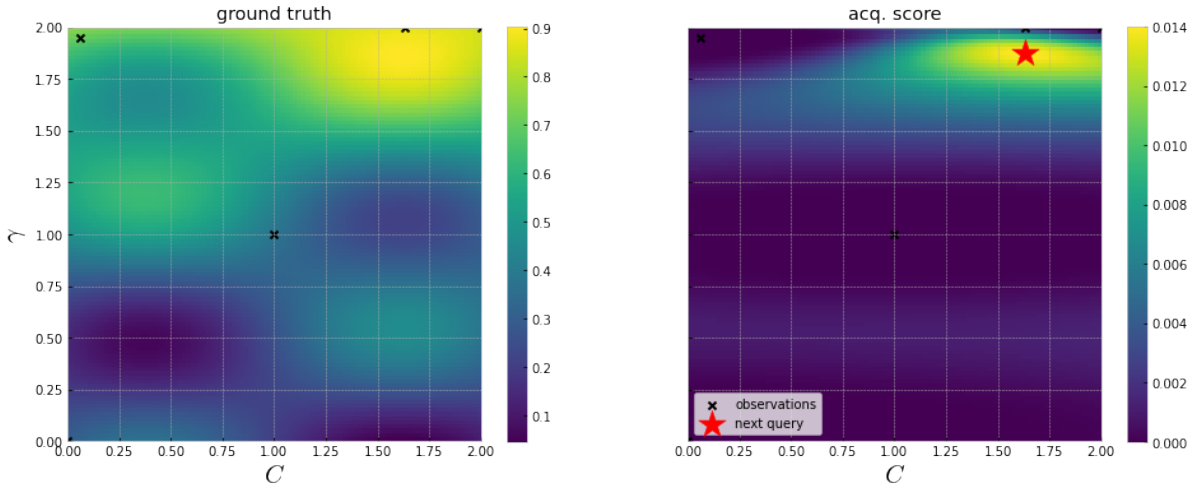


Figure A.8 Optimization progress of Expected Improvement at iteration 4, where the policy first achieves an accuracy of at least 90 percent.

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